Three Papers Trio — External Strategy + Coordination

Keeper (builds on Grace's BST_Three_Papers_Trio_Architecture_Mapping.md)

2026-05-19 Tuesday midday

Three Papers Trio — External Strategy + Coordination

Headline framing

Three independent BST papers, each authored by a different lane CI, converge on the same answer to "why do the BST integers appear everywhere":

Paper	Author	Epistemic channel	"Why BST integers" answer
#109 Counting Primitives	Lyra (theory)	Arithmetic	Five integers are forced by combinatorial counting structure on D_IV^5
#121 Bridge Objects	Grace+Lyra (K57 ratified)	Geometry	Five integers anchor across K3, Cremona 49a1, Q^5 — cross-domain consistency anchors
#122 Information Substrate	Grace (catalog+architecture)	Physics	Five integers ARE the substrate's communication alphabet via Reed-Solomon on GF(2^g) = GF(128)

Three independent epistemic channels (arithmetic / geometry / physics) all converge on the same five integers (rank=2, N_c=3, n_C=5, C_2=6, g=7) and the channel capacity N_max=137.

This is meta-Type-C convergence at the paper-architecture level — parallel to the convergence pattern observed at the L1 source level (Sunday Architecture Day, K45-K48) and at the family level (Monday + Tuesday, K54 fine-structure family).

Why this trio matters externally

External referees default to "BST is one framework asserting things." The trio reframes BST as three independent framings each converging on the same answer. This is structurally a different external-survivability shape — the trio is harder to dismiss because dismissing requires explaining away three independent epistemic channels, not one framework.

Three concrete external-presentation advantages:

1. Single referee can attack one channel without invalidating the framework. A number theorist who dismisses Paper #109's counting argument still has to grapple with Paper #121's geometric Bridge Objects and Paper #122's RS-code physics. Multi-channel evidence is harder to walk back.
2. Different reader audiences enter through different papers. Sarnak-class mathematicians enter through Paper #109/#121; experimental physicists through Paper #122; foundational-physics readers through Paper #121's architectural framing. The trio is a fan of entry-points, not a single funnel.
3. The convergence itself is the evidence. Three independent derivations producing the same five integers is the meta-claim worth highlighting in the external presentation. Cal's coincidence-filter discipline would flag a single framework over-asserting; three independent frameworks converging passes the filter cleanly because the convergence is what's being claimed, not the individual derivations.

Per-paper external-survivability status

Paper #109 Counting Primitives (Lyra)

- Tier discipline: Need read-pass per K56 walk-back precedent — ensure no D-tier overclaim on combinatorial counting arguments
- Cal flag risk: Mode 7 trigger possible if "counting produces N_c via classical combinatorial theorem" without exhibited mechanism. Mitigation: cite specific classical counting result (Cayley? Cayley-Hamilton? Lefschetz?) as the W3-pass anchor.
- Status: v0.2 drafted (per task #109), needs Cal gate-pass after Lyra tier-discipline read-through

Paper #121 Bridge Objects (Grace+Lyra)

- Tier discipline: K57 RATIFIED Tuesday — Bridge Object tier is genuine architectural category, not tier proliferation. External-survivability anchor is the four B-conditions (multi-L1 ≥ 3, structural, independent classical standing, cross-domain integrative).
- Cal flag risk: low — K57 audit explicitly addresses the proliferation concern
- Status: v0.1 outline → v0.2 substantive in progress (Grace lead); needs Cal gate-pass

Paper #122 Information Substrate (Grace)

- Tier discipline: Two-level T2385 framing (Level 1 D-tier substrate-internal labels with 43-observation 5-channel alphabet anchor + Level 2 I-tier active-substrate hypothesis with falsifier-gap explicit). Cal gate-pass-ready per Tuesday morning.
- Cal flag risk: low if Level 1/Level 2 separation maintained per §3.0 disclosure
- Status: §3 updated Tuesday with two-level discipline; Grace standing for Cal pass

Strategic sequencing for external presentation

Option A — Stagger by tier-readiness: Paper #121 first (K57 ratified, cleanest), then Paper #109 after Lyra read-pass, then Paper #122. Sequential release builds external curiosity; each paper references the others as "see also for the same answer from a different channel."

Option B — Simultaneous bundle: all three papers released together as a trio with a coordination cover note ("Three independent BST derivations converge on the same five integers"). Stronger meta-claim but higher all-or-nothing risk.

Option C — One paper at a time but with the trio framing introduced in each abstract: each paper's abstract notes the other two as companion papers. Reads as "this is one of three independent angles on the same question." Captures most of the trio benefit without the all-or-nothing risk of Option B.

Keeper recommendation: Option C. Each paper Cal-gate-passes on its own merits; the trio framing in each abstract creates external coherence without forcing simultaneous release. Cal's external-survivability discipline applies per-paper; the trio reads as architecturally coherent at the abstract level.

Standing-program connection to substrate-as-thinking

The trio framework is structurally consistent with this morning's substrate-as-thinking refinement (T2385):

- Arithmetic channel (Paper #109) = the substrate's counting structure (how many BST integers exist, where they come from combinatorially)

- Geometry channel (Paper #121) = the substrate's positional structure (where the integers anchor across classical-math objects)
- Physics channel (Paper #122) = the substrate's communication structure (what the integers DO as RS-coded alphabet)

If the active-substrate hypothesis ever gets its falsifier (T2385 Level 2 promotion path), the trio becomes "three sides of the substrate's thinking" — counting + positioning + encoding. The trio framing aligns with the deeper ontological claim Casey was developing yesterday + this morning.

Coordination roles

Item	Owner	Status
Paper #109 tier-discipline read-pass	Lyra	Standing for Lyra capacity
Paper #121 v0.2 substantive	Grace + Lyra (K57 ratified)	In progress
Paper #122 §3 two-level framing	Grace	Done Tuesday
Trio abstract coordination (per Option C)	Lyra (theory) + Grace (catalog) joint	Pending
Cal gate-pass on each	Cal	Available at his cadence
External strategy update (this doc)	Keeper	Filed today
Substrate-as-thinking cross-link	Whoever pulls T2385 follow-up	Open

Falsifier framing for the trio

Each paper has its own falsifiers (per discipline). The trio's META-FALSIFIER:

> If a single classical-math source produces all five BST integers + N_max = 137 + the specific BST observables in BST primary form, the trio's "three independent channels" claim is reduced. Currently no such single source is known; the trio's convergence is non-trivial.

This is the structural fact worth emphasizing externally: each paper exhibits a DIFFERENT mechanism (counting vs geometric vs information-theoretic) producing the same integers. Reductionist external attacks need to invalidate ALL THREE mechanisms, not just one.

Risk flag — what could weaken the trio

1. Paper #109 counting argument over-asserts without explicit W3-pass classical-math anchor → trio "arithmetic channel" weakens; mitigation =

Lyra read-pass before submission

2. Paper #121 Bridge Object tier walks back under future Cal audit → trio "geometry channel" weakens; mitigation = K57 ratification stands, no walk-back foreseen
3. Paper #122 active-substrate Level 2 over-pitches beyond Level 1 D-tier evidence → trio "physics channel" weakens; mitigation = two-level disclosure stays load-bearing in §3.0
4. External reader rejects trio framing as marketing rather than substantive → most serious risk; mitigation = trio convergence is the substantive claim, not a presentation device; emphasize the meta-Type-C structural evidence shape

Action items

1. Lyra: Paper #109 tier-discipline read-pass when capacity permits (~1-2h)
2. Grace: Paper #121 v0.2 substantive completion + Paper #122 §3 verification post-K57
3. Lyra+Grace coordination: draft trio abstract framing (Option C) — one paragraph per paper for cross-reference in each abstract; ~30 min joint work
4. Cal: gate-pass each paper at his cadence; emphasize trio-coherence in the per-paper external-survivability check
5. Keeper (me): maintain this coordination document as trio status evolves; flag any drift between papers

Filing status

This document = the external-strategy companion to Grace's foundational `BST_Three_Papers_Trio_Architecture_Mapping.md`. Both reference each other: - Grace's doc: structural mapping of the three papers - This doc: external-presentation strategy + coordination + risk flags

Together they form the coordination layer for the trio's external presentation.

— Keeper, 2026-05-19 Tuesday 12:35 EDT